

Website francescapanero.github.io

Work email francesca.panero@uniroma1.it; f.panero@lse.ac.uk

On maternity leave 12/2025 - 05/2026

Work experience

- May 2024 – **Assistant Professor (tenure-track), Università di Roma “La Sapienza”, Rome, Italy**
present Department of Methods and Models for Economics, Territory and Finance
- 2025 **2025 Visiting Lecturer, Lugano, Switzerland**
Department of Informatics, Università della Svizzera Italiana
- 2024 – **Visiting Fellow in Statistics and affiliate to the Grantham Institute on Climate**
present **Change and the Environment, London School of Economics, London, UK**
- Jul 2022 – **Assistant Professor (tenure-track), London School of Economics and Political**
May 2024 **Science, London, UK**
Department of Statistics. Also affiliate to the Data Science Institute
- Apr – Jun **Researcher, University of Oxford, Oxford, UK**
2022 Part-time researcher for the MPLS division on ED&I topics related to admission data
- Jul – Oct **Intern at J.P. Morgan Chase, London, UK**
2021 Associate Quantitative Researcher with expertise in applied ML/AI, trading team
- Mar – May **Visiting researcher at IDSIA, Lugano, Switzerland**
2021 Working with Dr Marco Scutari on fair ML models
- Winter 2020 **Visiting student at Duke University, Durham, North Carolina (USA)**

Education

- Oct 2017 – **PhD in Statistics, University of Oxford**
Jun 2022 Oxford-Warwick Statistical Programme
- Sep 2015 – **MSc in Stochastics and Data Science, University of Turin, Italy**
Jul 2017 Final grade: 110/110 *cum laude*
- Sep 2012 – **Allievi Honors Program, Collegio Carlo Alberto, Moncalieri, Italy**
Jul 2017 Diploma in Economics *with distinction* and MA in Statistics and Applied Mathematics *with distinction*
- Sep 2012 – **BSc in Mathematics, University of Turin, Italy**
Oct 2015 Final grade: 110/110 *cum laude*
- Sep 2007– **High School (with emphasis on Humanities), “V. Alfieri”, Turin, Italy**
Jul 2012 Final grade: 98/100

Teaching and supervision experience

Teaching qualification

- Jul 2024 **Postgraduate Certificate in Higher Education, AdvanceHE**

Teaching

- AY 2025/26 **Lecturer, Probability and Stochastic Processes**, *graduate. MEMOTEF Department, Sapienza*
- AY 2025/26 **Lecturer, Computational Tools for finance**, *graduate. MEMOTEF Department, Sapienza*
- Jul 2025 **Lecturer, Machine Learning in Practice, ME315**, *Summer school, LSE*
- AY 2024/25 **Lecturer, Deep Learning**, *PhD. MEMOTEF Department, Sapienza*
- AY 2024/25 **Lecturer, Probability and Stochastic Processes**, *graduate. MEMOTEF Department, Sapienza*
- Jul 2024 **Lecturer, Machine Learning in Practice, ME315**, *Summer school, LSE*
- AY 2023/24 **Lecturer, Deep Learning ST456**, *graduate. Department of Statistics, LSE*
- AY 2023/24 **Lecturer, Foundations of Machine Learning ST510**, *PhD. Department of Statistics, LSE*
- AY 2022/23 **Teaching Assistant, Deep Learning ST456**, *graduate. Department of Statistics, LSE*
- AY 2022/23 **Teaching Assistant, Artificial Intelligence ST449**, *graduate. Department of Statistics, LSE*
- 2018 – 2021 **Teaching Assistant, Applied Statistics**, *Department of Statistics, University of Oxford*
- 2019 **Tutor, Preliminary of Statistics and Data Science, Applied Statistics**, *Hertford College and Exeter College, University of Oxford*

Supervision

PhD supervision

- Fall 2025 **Emma Kopp**, *visiting from Paris Dauphine University*
- 2024 – present **Trevor Wroblewski and Xinhui Liu**, *PhD in Statistics, LSE*
joint with Dr K. Kalogeropoulos and Dr S. Geneletti

MSc supervision

- AY 2025/26 **Edoardo Antonini, Daniele D'Intino, Flavio Ionescu, Tiziano Lostuzzi**, *FINASS. Sapienza*
- AY 2023/24 **Trevor Wroblewski**, *MSc Statistics. Department of Statistics, LSE*
joint with Dr K. Kalogeropoulos and Dr S. Geneletti
- AY 2023/24 **Group of four students**, *MSc Data Science. Department of Statistics, LSE*
Capstone Project ST498, in collaboration with KPMG
- AY 2022/23 **Group of four students**, *MSc Data Science. Department of Statistics, LSE*
Capstone Project ST498, in collaboration with Deutsche Bank

Research production

Publications and preprints

- 2026 **The Role of Uncertainty in Assessing the Fairness of Machine Learning Models**
F. Panero, E. C. Wit, M. Scutari. *arXiv:2609.07959*
- 2026 **Filling survey gaps in food security monitoring with spatio-temporal additive Gaussian process models**
E. Kopp, S. Ishida, R. Legonye, F. Panero. *arXiv:2608.14314*

- 2025 **Dynamic sparse graphs with overlapping communities**
X. Miscouridou*, F. Panero*, A. Laos. *arXiv:2512.10717*
(*Joint first authors)
- 2025 **Hierarchical additive interaction modelling with Gaussian process prior and its efficient implementation for multidimensional grid data**
S. Ishida, F. Panero, W. Bergsma. *arXiv:2305.07073*
- 2025 **Mapping Student-GenAI Interactions onto Experiential Learning: The GENIAL Framework**
J. Cardoso Silva, D. Sallai, C. Kearney, F. Panero, M. Barreto. *SSRN 5674422*
- 2024 **Approach generative AI tools proactively or risk bypassing the learning process in higher education**
Sallai D., Cardoso-Silva J., Barreto M., Panero F., Berrada G., Luxmoore S. *LSE Public Policy Review*, 3(3)
- 2023 **On sparsity, power-law and clustering properties of graphex processes**
F. Caron, F. Panero, J. Rousseau. *Advances in Applied Probability*, 1-43
- 2022 **Achieving fairness with a simple ridge penalty**
M. Scutari, F. Panero, M. Proissl. *Statistics and Computing* 32, 77
- 2021 **Bayesian nonparametric disclosure risk assessment**
S. Favaro, F. Panero, T. Rigon. *Electron. J. Statist.* 15(2): 5626-5651
- 2021 **Optimal disclosure risk assessment**
F. Camerlenghi, S. Favaro, Z. Naulet, F. Panero. *The Annals of Statistics*, 49.2: 723-744
(All the authors contributed equally to the paper)
- [Refereed conference proceedings, discussions](#)
- 2026 **Enhancing modelling of food security through Bayesian inference**
Panero, F. and Ishida, S. *Proceedings of the Statistics and Data Science Workshop 2026, Caserta*
- 2023 **Issues with Sparse Spatial Random Graphs**
F. Panero. *Book of short papers CLADAG 2023*
- 2023 **A discussion on Evaluating Sensitivity to the Stick-Breaking Prior in Bayesian Nonparametrics**, by R. Giordano, R. Liu, M. I. Jordan, T. Broderick
X. Miscouridou, F. Panero. *Bayesian Analysis*
- 2019 **Issues with Nonparametric Disclosure Risk Assessment**
F. Camerlenghi, S. Favaro, Z. Naulet, F. Panero. *Book of short papers SIS 2019*
- [Working papers](#)
- Sparse spatial random graphs**
F. Panero, F. Caron, J. Rousseau

Talks and posters

- Jul 2026 **ISBA World Meeting, Nagoya, invited talk**
Dynamic sparse graphs with overlapping communities
- Jul 2026 **ISBA World Meeting, Nagoya, poster**
The Role of Bayesian Uncertainty in Assessing the Fairness of Machine Learning Models
- Mar 2026 **Statistics and Data Science conference, SIS, Caserta, invited talk**
Enhancing modelling of food security through Bayesian inference

- Oct 2025 **Migration and displacement action plan for Sub-Saharan Africa, Sapienza, seminar**
Predicting food security using secondary data, and a reflection on the links of food and migration
- Jun 2025 **Bayesian Nonparametrics 14, Los Angeles, invited talk**
Fast additive Gaussian Processes to model food security
- Jun 2025 **Conference of the Italian Statistical Society, Genoa, invited talk**
Interpretability and uncertainty quantification can enhance modelling of food security
- May 2025 **Giornate della Ricerca MEMOTEF, Sapienza University, seminar**
Can Bayes help food security estimation?
- May 2025 **Early Career NPStats workshop, LUISS, invited talk**
Can Gaussian Processes forecast food insecurity?
- May 2025 **Department of Computer Science, Università della Svizzera Italiana, seminar**
Can Gaussian processes forecast food security?
- Jan 2025 **STOR-i Annual Conference, Lancaster University, invited talk**
Forecasting food insecurity using Gaussian processes
- Jan 2025 **Department of Statistics, University of Warwick, seminar**
Forecasting food insecurity using Gaussian processes
- Dec 2024 **IMS ICSDS 2024, Nice, contributed talk**
Uncertainty in fair machine learning
- Sep 2024 **BIRS, Frontiers of Bayesian inference and data science, Oaxaca, invited talk**
Modeling sparse networks with Bayesian Nonparametrics
- Jul 2024 **Design and Analysis of Networked Experiments workshop, King's College London, invited talk**
Modeling sparse networks with Bayesian Nonparametrics
- Jul 2024 **ISBA World meeting, Venice, poster and invited talk**
Can Gaussian Processes forecast food insecurity?
- Jun 2024 **Satellite Workshop ISBA, Lugano, invited talk**
Modeling sparse networks with Bayesian Nonparametrics
- May 2024 **University of Southampton Statistical Sciences Research Institute, seminar**
Modeling sparse networks with Bayesian Nonparametrics
- Apr 2024 **Department of Statistics, University of Oxford, seminar**
Modeling sparse networks with Bayesian Nonparametrics
- Dec 2023 **IMS ICSDS 2023, Lisbon, contributed talk**
Achieving fairness with a simple ridge penalty
- Nov 2023 **School of Mathematics, University of Edinburgh, seminar**
Sparse spatial random graphs
- Nov 2023 **World Food Programme, Rome, seminar**
Forecasting food insecurities with Gaussian Processes
- Oct 2023 **Department of Statistics, University College London, reading group**
Achieving fairness with a simple ridge penalty
- Sep 2023 **Cladag 2023, Salerno, invited talk**
Sparse spatial random graphs
- Jun 2023 **Department of Mathematics and Statistics, Lancaster University, reading group**
Sparse spatial random graphs

- Jun 2023 **Statistics Research Showcase, LSE**, *seminar*
Sparse networks with Bayesian nonparametrics
- May 2023 **Department of Statistics, Università degli Studi di Padova**, *seminar*
Achieving fairness with a simple ridge penalty
- Mar 2023 **King's College London**, *seminar*
Sparse spatial random graphs
- Dec 2022 **CMStatistics**, *invited talk*
Sparse spatial random graphs
- Dec 2022 **Department of Statistics, University of Oxford**, *seminar*
Achieving fairness with a simple ridge penalty
- Nov 2022 **QuantumBlack, McKinsey**, *seminar*
Achieving fairness with a simple ridge penalty
- Jun 2022 **ISBA World Meeting**, *poster*
Bayesian nonparametric disclosure risk assessment
- Apr 2022 **BNP networking event**, *contributed talk*
Bayesian nonparametric disclosure risk assessment
- Dec 2021 **Department of Statistics, London School of Economics**, *seminar*
Sparse spatial random graphs
- Jul 2021 **ISBA World Meeting**, *contributed talk*
Sparse spatial random graphs
- Jun 2021 **Department of CS, University of the Italian Switzerland, Lugano**, *seminar*
Sparse spatial random graphs
- Apr 2021 **Young Statisticians Europe (YSE)**, *seminar*
Sparse spatial random graphs
- Mar 2021 **School of Mathematical Sciences, University of Nottingham**, *seminar*
Sparse spatial random graphs
- Dec 2020 **CMStatistics**, *contributed talk*
Sparse spatial random graphs
- Aug 2020 **Duke Graduate Summer Talks**
Sparse spatial random graphs
- Dec 2019 **CMStatistics**, *invited talk*
Optimal disclosure risk assessment
- Jul 2019 **Bocconi Summer School in Advanced Statistics and Probability, Como**, *poster*
Sparse spatial random graphs
- Jun 2019 **Bayesian Nonparametrics 2019, Oxford**, *poster*
Sparse spatial random graphs
- Jun 2019 **Italian Statistical Society 2019, Milan**, *invited talk*
Optimal disclosure risk assessment
- Mar 2019 **Workshop on Machine Learning, Amazon Development Centre Germany**, *poster*
Sparse spatial random graphs
- Jun 2018 **ISBA World Meeting, Edinburgh**, *poster*
Optimal disclosure risk assessment

Grants and Awards

- 2025 **Travel allowance**, *Visiting Lecturer, Department of Informatics, Università della Svizzera Italiana*
- Jun 2025 **Travel grant**, *Bayesian Nonparametric section, ISBA, Los Angeles*
- May 2025 **Sapienza research grant**, PI of the project “Bayesian models for sparse networks and applications” (€3000), RP1241910EF9AF1F
- Dec 2024 **Travel grant**, *IMS International Conference on Statistics and Data Science (ICSDS), Nice*
- Jul 2024 **Travel grant**, *International Society of Bayesian Analysis, Venice*
- Jun 2024 **Travel grant**, *ISBA satellite workshop, Lugano*
- Apr 2023 – **Research and Impact Support Fund**, *LSE*, Principal investigator of the project “Forecasting food insecurities with Gaussian Processes” (£20000). In collaboration with the United Nation World Food Programme.
- Apr 2024
- Jun 2022 **Best Poster Award**, *ISBA 2022*
- Jun 2022 **Travel grant**, *International Society of Bayesian Analysis, Montreal*
- Apr 2022 **Travel grant**, *Bayesian Nonparametrics Networking Event, Nicosia*
- Jul 2021 **Best Student/Postdoc Contributed Paper Award**, *ISBA 2021*
- Sep 2017 – **EPSRC full scholarship (4 years)**
present Scholarship covering salary, university fees and college fees for the entire length of the PhD.
- Sep 2012 – **Allievi Honors Program full scholarship, Collegio Carlo Alberto, Moncalieri, Italy**
Jul 2017 Scholarship covering university fees.

Service to profession

Organisation

- Jun 2026 Member of the scientific and organizing committees of BAYSM 2026, Nagoya
- 2026 j-ISBA Chair (junior section of the International Society for Bayesian Analysis)
- 2025 j-ISBA Chair-Elect
- 2022 – 2024 Co-organiser of Data Science Seminars, Department of Statistics, LSE.
- 2022 – 2024 Co-organiser of Women in Statistics, Department of Statistics, LSE.
- 2018 – 2022 Committee member of ED&I discussion group in the Department of Statistics, Oxford (2020-22). Co-organizer of the Oxford R Users Group (2018-19).

Referee and editorial service

Referee for: Artificial Intelligence and Statistics Conference; Computational Statistics & Data Analysis; Journal of Computational and Graphical Statistics; Journal of the American Statistical Association; Bayesian Analysis; Electronic Journal of Statistics; Fairness, Accountability, and Transparency Conference; Statistical Science.

Public engagement

- Feb 2025 Co-organiser of the j-SBA and Young African Statisticians Association Bayesian statistics webinar “A Journey Through Bayes: How Bayesian Statistics Improves Our Understanding of Cancer”
- Nov 2024 Organiser and discussant of the LSE Public Lecture “Data Visualisation: alive visual words”, with Federica Fragapane

- Mar 2022 **Organiser and moderator of the EquiStats event “Oxford women in Machine Learning”, University of Oxford**
- Jun 2021 **Organiser and co-speaker of the EquiStats event “Fairness in ML”, University of Oxford**
- 2018 – 2019 **Co-organiser R Users Group, Oxford**
- Jun 2018 **Oxford Maths Festival, Department of Mathematics, University of Oxford**

Skills

Languages

- Italian** Native speaker
- English** Fluent
- French** Intermediate
- Spanish** Beginner

Computer skills

Advanced: R, Python (NumPy, SciPy, Pandas, TensorFlow, Keras), $\text{\LaTeX}$

Intermediate: GitHub, MATLAB, Microsoft Windows and Office (ECDL)

Basic: HTML, SAS, SQL, C++, CUDA

Summer schools and workshops

- Nov 2021 **The Oxford Strategy Challenge**, *University of Oxford*
Consultancy challenge, working with a team on a start-up marketing and strategy project
- May – Jun 2021 **SeedWISE**, *University of Oxford*
Enterprise programme for women doing a PhD in STEM
- Jul 2019 **Bocconi Summer School in Advanced Statistics and Probability**, *Como, Italy*
Random Graphs and Complex Networks: Structure and Function. Prof. R. van der Hofstad (Eindhoven University of Technology, NL), Prof. S. Bhamidi (University of North Carolina, USA)
- Aug 2016 **Summer school “Applied Bayesian Statistics”**, *Como, Italy*
Bayes, Big Data, and the Internet.